Practice 1: Perimeter basics

Name: ____________________ Date: __________

Find the perimeter. Show your work.

1. A rectangle is 8 cm long and 3 cm wide. Find its perimeter.

Work space

2. A square has side length 6 m. Find its perimeter.

Work space

Practice 2: Missing side lengths

Find the perimeter or missing side length. Use equal side lengths when needed.

1. A rectangle is 14 ft long and 5 ft wide. Find its perimeter.

Work space

2. A square has perimeter 36 in. What is the length of one side?

Work space

Practice 3: Perimeter word problems

Write an equation and label your answer.

1. Maya walks around a rectangular garden that is 12 m by 9 m. How far does she walk in one lap?

Work space

2. A triangular sign has side lengths 7 in, 8 in, and 10 in. Find its perimeter.

Work space

Practice 4: Area of rectangles

Area measures the square units covering a surface. Find each area.

1. Calculate the area of a rectangle 11 cm long and 4 cm wide.

Work space

2. A classroom rug is 15 ft by 6 ft. Find its area.

Work space

Practice 5: Area of squares

Use area = side × side.

1. Find the area of a square with side length 9 yd.

Work space

2. A square tile has area 144 in^2. What is its side length?

Work space

Practice 6: Find a missing dimension

Use area = length × width.

1. A rectangle has area 72 cm^2 and length 9 cm. Find its width.

Work space

2. A rectangle has area 120 ft^2 and width 8 ft. Find its length.

Work space

Practice 7: Area word problems

1. A rectangular vegetable bed is 18 m long and 7 m wide. How many square meters does it cover?

Work space

2. Each poster is 2 ft by 3 ft. What area do 4 posters cover altogether?

Work space

Practice 8: Perimeter or area?

Decide which measurement is needed, then solve.

1. How much fencing is needed around a 20 yd by 13 yd dog pen?

Work space

2. How much carpet is needed to cover a 10 ft by 12 ft room?

Work space

Practice 9: Area and perimeter review

1. Find both the area and perimeter of a rectangle 16 cm by 5 cm.

Work space

2. Find both the area and perimeter of a square with side 7 m.

Work space

Practice 10: Composite rectangles

Break each shape into rectangles, then add the smaller areas.

1. An L-shape is made from a 10 cm × 4 cm rectangle and a 6 cm × 3 cm rectangle that do not overlap. Find its total area.

Work space

2. A floor has two sections: 8 ft × 5 ft and 4 ft × 3 ft. Find the total area.

Work space

Practice 11: Volume with unit cubes

Volume measures the space a solid occupies. For all volume problems in this workbook, assume the solids and containers are rectangular prisms. Find each volume.

1. A rectangular prism is 5 cubes long, 3 cubes wide, and 2 cubes high. How many unit cubes fill it?

Work space

2. A box is 7 units long, 2 units wide, and 4 units high. Find its volume in cubic units.

Work space

Practice 12: Volume formula

Use volume = length × width × height.

1. Find the volume of a prism measuring 9 cm × 4 cm × 3 cm.

Work space

2. Find the volume of a storage bin measuring 12 in × 5 in × 2 in.

Work space

Practice 13: Missing volume dimension

1. A prism has volume 96 m^3, length 8 m, and width 4 m. Find its height.

Work space

2. A prism has volume 180 ft^3, width 6 ft, and height 5 ft. Find its length.

Work space

Practice 14: Volume word problems

1. An aquarium is 10 dm long, 4 dm wide, and 5 dm high. What is its volume in cubic decimeters?

Work space

2. A shipping box is 8 in × 6 in × 5 in. What is its volume?

Work space

Practice 15: Volume and layers

Each cube on this page is a unit cube. Think of volume as the number of cubes in one layer times the number of layers.

1. A prism has a base of 6 cubes by 4 cubes and 3 layers. Find its volume.

Work space

2. A box has 48 cubes in each layer and 5 layers. How many cubes fit inside?

Work space

Practice 16: Units and conversions

Use 1 foot = 12 inches and 1 meter = 100 centimeters.

1. A rectangle is 3 ft long and 2 ft wide. Find its area in square feet.

Work space

2. A rectangle is 200 cm long and 3 m wide. Convert the length to meters, then find its area in square meters.

Work space

Practice 17: Mixed measurement reasoning

1. A rectangular garden is 9 m by 6 m. Its fence has a gate opening 2 m wide and 1 m high. How many meters of fence are needed, excluding the gate opening?

Work space

2. A box is 4 cm on each edge. Find its volume and explain why the unit is cubic centimeters.

Work space

Practice 18: Multi-step applications

1. A rectangular playground is 25 m by 14 m. A 3 m by 4 m sandbox occupies part of it. What area remains for play?

Work space

2. A 12 in by 10 in by 6 in box is filled with cubes. Each cube is 1 cubic inch. How many cubes are needed?

Work space

Practice 19: Explain your strategy

Show equations and words.

1. Two rectangles have the same area of 48 cm^2. One is 12 cm by 4 cm. Give another whole-number pair of side lengths and compare their perimeters.

Work space

2. A prism has dimensions 3 ft, 4 ft, and 5 ft. If the length doubles, what is the new volume? Explain.

Work space

Practice 20: Final review

Choose the correct measurement and show all work.

1. A square patio has side length 11 m. Find its perimeter and area.

Work space

2. A rectangular prism is 15 cm long, 4 cm wide, and 3 cm high. Find its volume. Then find the area of its rectangular top.

Work space